Firmware Flashing & Setup Guide

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Important safety note
Do not power the LED boards from the USB programming port. Use the correct external LED power supply for the PCB/LED assembly and keep the ESP32 ground and LED power ground common. Do not connect raw vehicle signals directly to ESP32 GPIO pins; use the PCB signal-conditioning circuit only.

1. Purpose and Scope

This document explains how to prepare a fresh computer and flash the Tail Light firmware onto the ESP32-S3 based PCB. It is intended for firmware re-installation, future maintenance, and basic verification after programming.

2. Firmware and Hardware Overview

The firmware controls a four-board automotive tail light assembly using an ESP32-S3 N16R8 controller. It drives four SK6812 RGBW LED PCBs, handles brake/turn/hazard priority states, and supports BLE-controlled show animations when vehicle signal inputs are idle.

Item	Project Value
Controller	ESP32-S3 N16R8
LED Type	SK6812 RGBW addressable LEDs
LED Boards	Four boards: L1, L2, R1, R2
LED Count	78 RGBW LEDs per PCB
Firmware File	Tail_Light_Firmware.ino
Custom Flash Layout	partitions.csv for 16 MB flash
BLE Device Name	Tail Light
Serial Monitor Baud Rate	115200

Operating priority
Vehicle lighting states override decorative BLE animations. The firmware priority order is: hazard + brake, hazard, turn + brake, turn, brake, BLE animation, then normal running light.

3. Required Software and Resources

Resource	Purpose	Link / Source
Arduino IDE 2.x	Compile and upload the sketch	https://www.arduino.cc/en/software
ESP32 Arduino Core	Adds ESP32-S3 board support to Arduino IDE	https://espressif.github.io/arduino-esp32/package_esp32_index.json
Adafruit NeoPixel Library	Required LED driver library for SK6812 RGBW LEDs	Install from Arduino Library Manager

4. Downloading and Preparing the Firmware Folder

Extract the shared Fimrware files in any storage partition for example D drive.
Place the extracted folder inside the Arduino sketchbook folder, for example:

D/Tail_Light_Firmware

Confirm that the folder contains Tail_Light_Firmware.ino and partitions.csv in the same top-level folder.
Open Tail_Light_Firmware.ino from Arduino IDE. Do not copy only the .ino file by itself; keep the complete project folder together.

```
Tail_Light_Firmware/
|-- Tail_Light_Firmware.ino
|-- partitions.csv
|-- Docs/
```

5. Installing Arduino IDE

Go to the official Arduino software download page: https://www.arduino.cc/en/software
Download Arduino IDE 2.x for the operating system being used: Windows, macOS, or Linux.
Install Arduino IDE using the default installer options.
Launch Arduino IDE once after installation so it creates the default sketchbook folder.

6. Installing ESP32-S3 Board Support

Open Arduino IDE.
Go to File > Preferences.
In Additional boards manager URLs, add the stable Espressif package URL:

```
https://espressif.github.io/arduino-esp32/package_esp32_index.json
```

Click OK.
Open Tools > Board > Boards Manager.
Search for esp32 and install esp32 by Espressif Systems.
Restart Arduino IDE after the board package is installed.

> **Recommended package selection**
> Use the latest stable ESP32 package unless a locked release is provided with the project. Avoid development/nightly board packages unless specifically required.

7. Installing Required Libraries

The firmware uses one external Arduino library and several ESP32-core libraries. Install only the external library manually; the rest come with the ESP32 board package.

Library / Header	Required Action	Reason
Adafruit_NeoPixel.h	Install from Arduino Library Manager	Controls SK6812 RGBW LED strips
BLEDevice.h	No manual install	Included with ESP32 Arduino core
Preferences.h	No manual install	Included with ESP32 Arduino core for NVS storage
esp_idf_version.h / esp_task_wdt.h	No manual install	Included through ESP32/ESP-IDF toolchain
freertos/FreeRTOS.h / freertos/queue.h	No manual install	Included through ESP32 Arduino core

Install Adafruit NeoPixel

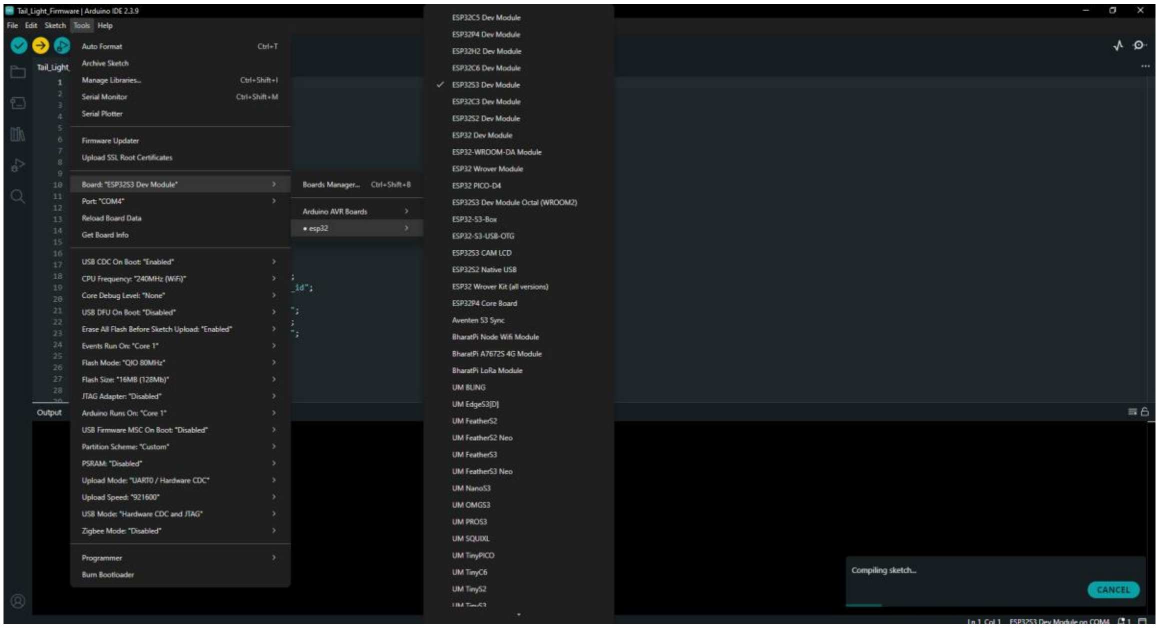
Open Arduino IDE.
Go to Sketch > Include Library > Manage Libraries.
Search: Adafruit NeoPixel.
Install the library named Adafruit NeoPixel by Adafruit.
After installation, close and reopen Arduino IDE if the library is not detected immediately.

Avoid conflicting BLE libraries
Do not install ArduinoBLE for this project. The firmware uses ESP32 BLEDevice.h from the ESP32 Arduino core, not the generic ArduinoBLE library.

8. Recommended Arduino IDE Board Settings

Connect the PCB to the computer using a known-good USB data cable, then configure Arduino IDE before compiling or uploading.

Arduino IDE Menu	Recommended Setting	Notes
Tools > Board	ESP32S3 Dev Module	Use this for a custom ESP32-S3 PCB unless a dedicated board profile is supplied.
Tools > Port	Detected COM/USB port	Port appears only after the PCB is connected and drivers are working.
Tools > USB CDC On Boot	Enabled it its important	It’s important that you enable this because without you won’t be able to flash properly
Tools > CPU Frequency	240 MHz	Default ESP32-S3 performance setting.
Tools > Flash Size	16MB	Required for ESP32-S3 N16R8 and project partition layout.
Tools > PSRAM	Leave it as it is	N16R8 includes PSRAM; firmware does not heavily depend on it.
Tools > Partition Scheme	Custom / project partitions.csv	Keep partitions.csv in the sketch folder. Use Custom if shown by the ESP32 core.
Tools > Upload Speed	921600 or 460800	Use 460800 if upload is unstable.
Tools > Core Debug Level	None	Use default for client flashing.
Tools > Erase All Flash Before Sketch Upload	Enabled for first install, Disabled for later updates	First flash erase clears old NVS/settings and avoids stale behavior.



Make sure that highlighted options are properly set

> **Partition scheme note**
> The repository includes partitions.csv for a 16 MB flash layout. Keep this file beside Tail_Light_Firmware.ino. If the Arduino IDE does not show a Custom partition option, update the ESP32 board package or use an available 16 MB partition profile only after confirming that the sketch compiles and uploads correctly.

9. PCB Connection and Safety Checks

Check	Requirement
USB cable	Must be a data-capable USB cable, not charge-only.
PCB power	Use correct PCB input supply as defined by hardware design.
LED power	Use external LED power supply sized for LED current. Do not power LEDs from USB.
Ground reference	ESP32 ground, LED supply ground, and signal-conditioning ground must be common.
Vehicle inputs	Brake, left turn, and right turn must pass through the PCB conditioning stage.
Signal level	ESP32 GPIO expects safe logic-level input. Idle should be LOW and active should be HIGH.
USB port	Connect to the ESP32 programming/USB-UART port on the PCB.
BOOT/EN buttons	Locate BOOT and RESET/EN pads/buttons before flashing in case manual bootloader entry is needed.

10. Flashing the Firmware

Open Tail_Light_Firmware.ino in Arduino IDE.
Confirm the correct board and settings from Section 8.
Connect the PCB to the PC using USB.
Select the correct serial port from Tools > Port.
Click Verify/Compile. Resolve any library or board errors before uploading.

Click Upload.
If the IDE shows connecting... for a long time or fails to enter bootloader, use manual boot mode: hold BOOT, press and release EN/RESET, then release BOOT after upload starts.
Wait until Arduino IDE reports done uploading.
Press EN/RESET or power-cycle the PCB after upload.

First installation recommendation
For the first flash, enable Erase All Flash Before Sketch Upload. This removes old NVS values and guarantees the running-light mode and BLE settings start cleanly.

11. First Power-On Verification

Verification Step	Expected Result
Serial Monitor	Set baud rate to 115200. Firmware prints: Tail light Arduino firmware started.
Startup LED Check	LEDs clear, run the startup color/light check, then switch to the running-light state.
Running State	With brake and turn inputs idle, LEDs show the configured red running light mode.
Brake Input	Brake signal forces all four boards to full brake red.
Left Turn Input	L1 and L2 animate together in amber; right side remains in base state unless brake/hazard is active.
Right Turn Input	R1 and R2 animate together in amber; left side remains in base state unless brake/hazard is active.
Hazard Input	Left and right activity together flashes all boards’ amber.
BLE Scan	Mobile BLE scanner should show device name Tail Light when signals are idle.

12. BLE Command Test

You can use the mobile application to test the LED animations

13. Troubleshooting

Problem	Likely Cause	Fix
Board/COM port not visible	USB cable is charge-only, driver missing, or wrong USB port	Use a data cable, try another USB port, install CP210x/CH340 driver if the PCB uses that chip.
Upload fails at Connecting...	Auto-reset circuit did not enter bootloader	Hold BOOT, press/release EN or RESET, and then release BOOT after upload begins.
Compilation error: Adafruit_NeoPixel.h not found	NeoPixel library not installed	Install Adafruit NeoPixel from Library Manager and restart Arduino IDE.
Compilation error: BLEDevice.h not found	ESP32 board package not installed/selected	Install esp32 by Espressif Systems and select ESP32S3 Dev Module.
Upload succeeds but LEDs do not turn on	No external LED power, missing common ground, wrong LED data connection, or LED supply issue	Verify external LED supply, common GND, and data GPIO mapping.
Only USB works but external power fails	Board power path or regulator issue	Check PCB input voltage, regulator output, and ESP32 3.3 V rail.
BLE device not visible	Vehicle input active, BLE not initialized, phone cache issue	Set all inputs idle, reset PCB, rescan BLE, or toggle phone Bluetooth.

Problem	Likely Cause	Fix
Wrong colors or flickering LEDs	Incorrect LED type/order or unstable power	Confirm SK6812 RGBW/GRBW configuration, short data wire, stable 5 V LED supply, and common GND.
Unexpected saved running mode	Old NVS settings retained	Enable Erase All Flash Before Sketch Upload once, then re-upload.

14. Final Client Checklist

- Arduino IDE installed successfully.
- ESP32 board package installed using the stable Espressif board manager URL.
- ESP32S3 Dev Module selected in Arduino IDE.
- Flash size set to 16MB.
- Partitions.csv kept with Tail_Light_Firmware.ino.
- Adafruit NeoPixel library installed.
- PCB detected on the correct COM/USB port.
- Firmware compiles without errors.
- Firmware uploads successfully.
- Serial monitor prints firmware start message at 115200 baud.
- Running, brake, turn, hazard, and BLE modes verified.

Appendix A: Pin Map and Partition Table

A.1 Firmware Pin Map

Function	GPIO
Brake input	GPIO 1
Left turn input	GPIO 2
Right turn input	GPIO 38
LED strip L1	GPIO 4
LED strip L2	GPIO 16
LED strip R1	GPIO 12
LED strip R2	GPIO 14
Ready status LED	GPIO 17
Heartbeat status LED	GPIO 8

A.2 Custom 16 MB Partition Table

Name	Type	Subtype	Offset	Size
nvs	data	nvs	0x9000	0x5000
otadata	data	ota	0xE000	0x2000
app0	app	ota_0	0x10000	0x640000
app1	app	ota_1	0x650000	0x640000
spiffs	data	spiffs	0xC90000	0x370000

Appendix B: Useful Links

Arduino IDE Download: https://www.arduino.cc/en/software
ESP32 Arduino Core Installation Guide: https://docs.espressif.com/projects/arduino-esp32/en/latest/installing.html
ESP32 Stable Board Manager URL: https://espressif.github.io/arduino-esp32/package_esp32_index.json
Adafruit NeoPixel Library: https://github.com/adafruit/Adafruit_NeoPixel